Pairs of symmetric parabolics and their full graph groups

Human Compression Project

Abstract

For prime $p \equiv \pm 1 \pmod 8$, the normalizer of a dihedral subgroup of order eight determines the two S_4 parabolics above it. According to one parity, these parabolics belong to the same conjugacy class or to different classes. This distinction gives the full graph groups $\mathrm{PSL}(2,p) \times C_2$ and $\mathrm{PGL}(2,p)$, respectively. The proof uses a concrete type interchange and Tutte's cubic stabilizer bound; it requires no graph automorphism search. The source's conjugacy counts are recovered, while four of its nine prime-field full-group identifications require correction.

1 The normalizer determines the parabolics

Let $p \geq 7$ be prime with $p \equiv \pm 1 \pmod 8$. Choose $\epsilon \in \{1, -1\}$ so that $p \equiv \epsilon \pmod 8$, and put

$$G = \mathrm{PSL}(2,p), \quad E = \mathrm{PGL}(2,p), \quad n = \frac{p-\epsilon}{2}, \quad d = \frac{p-\epsilon}{8}.$$

Thus $n = 4d$. We retain the classical subgroup facts used by the source: the maximal subgroups S_4 of G form two conjugacy classes, fused by E; their Sylow two-subgroups are D_8; cyclic groups of order four lie in a unique torus of order n in G and $2n$ in E; and those torus normalizers are dihedral. Also G is simple and $\mathrm{Aut}(G) = E$. The cyclic torus containing an element of order four is its projective centralizer, so its uniqueness here is literal, not just an abstract order identification.

Lemma 1. *For $B \leq G$ isomorphic to D_8,*

$$\mathrm{N}_E(B) \cong D_{16}, \qquad \mathrm{N}_G(B) = \begin{cases} D_{16}, & d \text{ even}, \\ B, & d \text{ odd}. \end{cases}$$

The subgroups B form one E-class. This class splits into two G-classes when d is even and remains one class when d is odd.

Proof. The cyclic subgroup C_4 is characteristic in B, so every normalizer of B normalizes its unique torus. Write the torus normalizer in E as

$$\langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \qquad \mathrm{N}_G(C_4) = \langle R^2, S \rangle,$$

choosing a reflection $S \in G$. After conjugacy in this normalizer we may take $B = \langle R^{n/2}, S \rangle$. Indeed, its reflections in G have even exponents $R^{2j}S$, and conjugation by R^j is transitive on these possibilities modulo the rotation subgroup of B. Together with conjugacy of the tori this proves that the D_8 subgroups in G form one E-class.

A rotation R^k normalizes B exactly when $(n/2) \mid 2k$, or $d \mid k$. Its normalizing rotations are therefore $\langle R^d \rangle$, of order eight. The full normalizer is $\langle R^d, S \rangle \cong D_{16}$. Since $R^k \in G$ exactly

when k is even, this normalizer lies in G when d is even. When d is odd, its intersection with G is $\langle R^{2d}, S\rangle = B$. For a subgroup of index two, an E-conjugacy class splits into two G-classes precisely when its normalizer is contained in G: this follows directly by comparing the G-orbit and E-orbit lengths. This proves the final assertion. □

Lemma 2. *Each $B \cong D_8$ is contained in exactly two subgroups M_0, M_1 isomorphic to S_4. Their intersection is B. If d is even, they belong to the same G-class and are interchanged by $\mathrm{N}_G(B)/B$. If d is odd, they belong to different G-classes and are interchanged by $\mathrm{N}_E(B)/B$. In either case there is an element t interchanging them by conjugation with $t^2 \in B$; one may take $t \in G$ exactly in the even case.*

Proof. Every S_4 contains one conjugacy class of three subgroups D_8, and $\mathrm{N}_{S_4}(D_8) = D_8$. Fix an S_4 class containing the given B. Its members containing B form a transitive $\mathrm{N}_G(B)$-set: adjust a transporter inside S_4 so that it also preserves B. The stabilizer of one member is $\mathrm{N}_G(B) \cap \mathrm{N}_G(S_4) = B$, since a maximal S_4 is self-normalizing in the simple group G. Thus the number of such members in this class is $[\mathrm{N}_G(B) : B]$.

For even d, the two B-classes are exchanged by the outer coset, as are the two S_4 classes. Each S_4 class contains only one B-class; the two S_4 classes consequently contain different B-classes. A fixed B therefore has two overgroups in one class and none in the other. Their interchange is the regular action of $\mathrm{N}_G(B)/B \cong C_2$.

For odd d, there is just one B-class. Each of the two S_4 classes supplies one overgroup. Every element of $\mathrm{N}_E(B) \setminus G$ interchanges these overgroups, because it exchanges their G-classes. In both cases the relevant normalizer is D_{16} with B of index two, so the square of an interchanging element belongs to B. Finally B is maximal in S_4; distinct equal-order overgroups containing it therefore meet exactly in B. □

These pairs give connected cubic bipartite coset graphs. The local action of S_4 on its three cosets of D_8 is S_3, and distinct maximal parabolics generate G. Thus all the rank two geometry and local two-transitivity conditions hold.

Theorem 3. *Under the source's unordered convention, the configurations with parabolics S_4, S_4 and intersection D_8 have two conjugacy classes when d is even and one when d is odd. In both cases there is one class under* $\mathrm{Aut}(G)$.

Proof. For each B the unordered pair of overgroups is unique, by Lemma 2. Its intersection recovers B. Conjugacy and automorphism classes of pairs therefore correspond exactly to the respective classes of B counted in Lemma 1. This proves the result, including the type interchange identification. □

2 A literal type interchange

Use left cosets gM_i as vertices of type i, with incidence defined by nonempty intersection. Left multiplication gives the specified type-preserving action of G. Let t be as in Lemma 2 and write $\alpha(g) = tgt^{-1}$.

Lemma 4. *The map*

$$T(gM_i) = \alpha(g)M_{1-i}$$

is a graph automorphism interchanging the two types. It satisfies $T^2 = L_{t^2}$, where L_g denotes left multiplication by $g \in G$. If $t \in G$, then $C = L_{t^{-1}}T$ is an involution centralizing the whole action of G.

Proof. Conjugation by t interchanges the two parabolics, so the displayed formula is independent of the coset representative. It carries intersecting cosets to intersecting cosets and is bijective, so it is a graph automorphism. Since $t^2 \in B \subseteq M_i$,

$$T^2(gM_i) = t^2gt^{-2}M_i = t^2gM_i = L_{t^2}(gM_i).$$

When $t \in G$, the adjusted map has the simpler expression $C(gM_i) = gt^{-1}M_{1-i}$. It commutes with every left multiplication, and its square is $gM_i \mapsto gt^{-2}M_i = gM_i$. It is nonidentity because it exchanges the two types. □

3 The full automorphism group

We retain Tutte's theorem for finite connected simple cubic arc-transitive graphs: the full vertex stabilizer has order at most 48. The source and the separate subdivision proof state this classical premise with its original references. Our graph is simple by its coset-incidence definition and connected by maximality. Lemma 4, together with the local action of G, gives arc transitivity of its full automorphism group.

Theorem 5. *The full automorphism group A of this incidence graph is*

$$A \cong \begin{cases} \mathrm{PSL}(2,p) \times C_2, & d \text{ even}, \\ \mathrm{PGL}(2,p), & d \text{ odd}. \end{cases}$$

In both cases its exact arc-transitivity length is four.

Proof. There are $2|G|/24$ vertices and A is vertex transitive. Tutte's bound gives

$$[A : G] = \frac{2|A_x|}{24} \leq 4.$$

The kernel of the action of A on its cosets of G is a normal subgroup of A contained in G, hence a normal subgroup of the simple group G. If this kernel were trivial, G would embed in S_4, impossible since $|G| \geq 168$. Therefore the kernel is G, and $G \triangleleft A$.

An element of $\mathrm{C}_A(G)$ either preserves both G-orbits of vertices or exchanges them. If it preserves both, it is the identity on each: the centralizer of a transitive coset action G/M_i is $\mathrm{N}_G(M_i)/M_i = 1$. This equality also follows directly by sending the base vertex to another vertex fixed by its stabilizer. Thus the centralizer has order at most two. If M_0, M_1 are in different G-classes, it cannot exchange the orbits, because a G-equivariant bijection of transitive G-sets requires conjugate point stabilizers.

For even d, Lemma 4 supplies precisely the nontrivial centralizer element C. Conjugation embeds $A/\langle C\rangle$ into $\mathrm{Aut}(G) = E$. Its image preserves the single G-class containing both vertex stabilizers. Every outer automorphism exchanges the two S_4 classes, so this image is $\mathrm{Inn}(G)$. Since G is centerless, it intersects $\langle C\rangle$ trivially, and consequently $A = G \times \langle C\rangle$.

For odd d, the centralizer is trivial and conjugation embeds A in E. The constructed type interchange T is outside the given type-preserving G. Therefore $|A| \geq 2|G| = |E|$, proving equality with E. This proves both full-group assertions.

In either case $|A| = 2|G|$, so $|A_x| = 24 = |G_x|$. The previously proved local length four for the specified S_4, S_4, D_8 action therefore remains the exact length for A: it has the same vertex stabilizers. The type interchange makes this a global four-arc transitive action. A fifth arc would require a vertex stabilizer divisible by $3 \cdot 2^4 = 48$, which is impossible. □

4 Comparison with the fixed source

Theorem 3 recovers Table 3, rows 7–8, including its unordered convention. Theorem 5 applies to the nine prime fields listed on printed page 386 of the fixed source, as follows.

p	d	Full graph group	Printed identification
17	2	$\mathrm{PSL}(2,17) \times C_2$	corrected
23	3	$\mathrm{PGL}(2,23)$	recovered
31	4	$\mathrm{PSL}(2,31) \times C_2$	corrected
41	5	$\mathrm{PGL}(2,41)$	recovered
47	6	$\mathrm{PSL}(2,47) \times C_2$	corrected
71	9	$\mathrm{PGL}(2,71)$	recovered
73	9	$\mathrm{PGL}(2,73)$	recovered
79	10	$\mathrm{PSL}(2,79) \times C_2$	corrected
89	11	$\mathrm{PGL}(2,89)$	recovered

The source identifies all nine full groups as $\mathrm{PGL}(2,p)$. In the four corrected cases our explicit central involution already disproves that identification: $\mathrm{PGL}(2,p)$ has trivial center, since its centralizer of the normal simple subgroup $\mathrm{PSL}(2,p)$ is trivial. The upper-bound argument proves the exact replacement, not just the failure of the printed name. There is no second principal-refutation credit: this is a surviving-claim correction in an original already registered as principally refuted. The prime-square case $q = 9$ is separate and is not covered by this proof.

The geometry family and its source counts are attributed to the original authors and the earlier work cited there. We retain the classical subgroup structure, outer action and Tutte's bound. The contribution here is the parity-controlled normalizer argument, the explicit centralizing type interchange and the resulting corrected full group theorem. The project root supplied the common Tutte/core/ centralizer upper-bound route during collaboration; the authoring agent supplied the normalizer parity, overgroup analysis and explicit type maps. The combined proof requires distinct review. It claims no whole-source completion, new open-conjecture answer, Lean verification or compression measurement.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2,q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Table 3 and p. 386.

[2] W. T. Tutte, *A family of cubical graphs*, Proc. Cambridge Philos. Soc. **43** (1947), 459–474.

[3] W. T. Tutte, *On the symmetry of cubic graphs*, Canadian J. Math. **11** (1959), 621–624.